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RESEARCH ARTICLE

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Investigating adverse daily life effects following a psychosocial laboratory stress task, and the moderating role of Psychopathology

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ABSTRACT

Laboratory stress tasks are necessary to closely investigate the stress response in a controlled environment. However, to our knowledge, no study has tested whether participating in such tasks can pose any daily life adverse effect. Fifty-three healthy participants (46 women) took part in a laboratory session where stress was induced using a typical psychosocial stressor: the repeated Montreal Imaging Stress Task (rMIST). Average levels of negative affect (NA), heart rate (HR), root mean square of successive differences (RMSSD), and skin conductance level (SCL), as well as reactivity across all these parameters as measured with the experience sampling method (ESM) in the four days prior to the laboratory session were compared with the four days following the session. We also assessed whether vulnerability to psychopathology moderated these associations. Findings showed that the task did not pose any significant adverse effect on participants. However, there was an unexpected increase in average RMSSD and a decrease in average SCL pre- to post-task. In addition, more vulnerable individuals were more likely to experience an increase in average levels of NA in the days following the task compared to the days preceding it. Our findings suggest that laboratory stress tasks may pose a significant risk to more vulnerable individuals.

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Stress; Laboratory; vulnerability; ESM; reactivity; psychopathology

Introduction

Stress is a leading health complaint driving a wide range of psychopathological health outcomes (Turner et al., 2020). Researching stress in the wild presents several complex challenges. Most notably, participants scarcely report on stress in the moment that it occurs. Instead, responses are often delayed, making it difficult to draw reliable insights (De Calheiros Velozo et al., 2023). Thereby, psychosocial laboratory stress tasks provide a controlled environment where stress is elicited in order to closely investigate key markers of the response (Bali & Jaggi, 2015) and their role in the development of mental illness (Ellenbogen et al., 2006; Morris et al., 2014; Pruessner et al., 2013; Yoon & Joormann, 2012). Although these tasks are designed to induce only mild levels of stress, and as such their effects on daily life are expected to remain limited, there is, to our knowledge, no study actively investigating the repercussions of such a stress protocol. Therefore, the purpose of this study is to assess whether inducing stress in the laboratory can in fact elicit any adverse effect on the participant that lingers beyond the task itself and into the days that follow it.

Psychosocial stress tasks rely on known elements of stress to elicit a response, namely: novelty, uncontrollability, and social evaluative threat (Dickerson & Kemeny, 2004). These tasks have shown to successfully elicit a significant stress response on an affective, as well as a physiological level (Allen et al., 2014; Giles et al., 2014; Hellhammer & Schubert, 2012). Specifically, there is an affective response marked by an increase in negative affect (NA) (Boesch et al., 2014; Brown et al., 2012). Likewise, task-induced changes in sympathetic and parasympathetic activity are reflected in increases in measures of heart rate (HR) and skin conductance level (SCL), as well as a decrease in heart rate variability measured as the root mean square of successive differences (RMSSD) (Kamarck & Lovallo, 2003; Ulrich-Lai & Herman, 2009; Visnovcova et al., 2013). Although these responses reflect adaptive behavior and are crucially necessary to maintain allostasis, prolonged activation can result in sensitization of the stress system, whereby subsequent stressors trigger an increased reactivity that further sensitizes the system with potentially detrimental consequences (Harkness et al., 2015).

Daily life studies conducted with the experience sampling method (ESM), have shown that stress reactivity is exacerbated in individuals who are more sensitive to stress, such as those at an increased clinical risk to develop a mental disorder (Bale, 2006; Guidi et al., 2021; Krkovic et al., 2018). Experience

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sampling is a diary method where participants complete a questionnaire multiple times a day for several days to capture changes within individuals in their natural environment as they go about their life (Myin-Germeys et al., 2018; Palmier-Claus et al., 2019; van Berkel et al., 2018). Here, findings consistently show aberrant stress responses in individuals with or at risk of a mental illness (e.g. depression, psychosis, bipolar disorder) when compared to healthy controls (Myin-Germeys et al., 2018). Given the fact that exposure to stress has the potential to further sensitize the stress system, it beckons the question of whether exposure to a psychosocial stress task may in itself yield a significant daily life adverse effect on participants, especially in those who are already at a more vulnerable state.

In the current study, we aim to investigate this very question. Namely, whether participating in a laboratory stress task poses an adverse effect beyond the scope of the experiment, that is in the days following stress induction. We combine ESM with wearable devices to capture key markers of the affective and physiological stress response in daily life (Brannon et al., 2016). In this way, it is possible to compare if there are any meaningful daily life changes following the laboratory stress induction compared to the days preceding it. We conclude that the task should pose no daily life adverse effect if (1) there is no significant increase in average level of NA, HR, and SCL, or decrease in RMSSD, from the four days pre- to the four days post- task; and likewise (2) there is no stress sensitization, that is no significant increase in stress reactivity pre- to post- task. Moreover, in light of the fact that vulnerable individuals are more stress sensitize, it is possible that psychopathology may increase the risk of experiencing an adverse effect from the task.

Methods

Sample

Fifty-three volunteers between the ages of 18 and 35 were recruited from the general population with the use of flyers distributed throughout the city and shared online. A power calculation was performed prior to recruitment to ensure sufficient data for the analyses planned. A sample of 50 participants was deemed sufficient. Participants were excluded if they had any cardiovascular and/or endocrinological disorders. Likewise, participants who had any relevant skin allergies were also excluded as the wearable devices, especially the Stingray, which is a chest patch used to collect continuous measures of HR and HRV, can be quite harsh on sensitive skins over prolonged use. The current study was part of a larger study, the *Study on Mental effort In the Lab and in daily Life* (SMILE: De Calheiros Velozo et al., 2021). It was advertised as focusing on mental effort and as such all participants were given this cover story to distract from the real purpose of the study (i.e. stress). Participants received a compensation of up to100€(depending on their compliance to the ESM protocol) for their participation in all parts of the study. It was approved by the Social and Societal Ethics Committee (SMEC) from KU Leuven, with all participants signing an informed consent prior to their participation.

Procedure

Protocol

Participants were first invited to an information session where they were carefully briefed about the study and what was expected of them. They completed questionnaires measuring demographic factors (e.g. age, gender, education, etc.) and psychopathology. In addition, participants were given a study phone to fill-out the ESM assessments, and a set of wearables to measure their physiology. Participants then all took part in eight days of ESM where measures of physiology were also collected. Halfway through the ESM period (i.e. end of fourth day), participants participated in a laboratory session where stress was elicited with the repeated Montreal Imaging Stress Task (rMIST).

Repeated Montreal Imaging Stress test (rMIST)

Stress was induced in the laboratory with the rMIST, a slightly modified version of the MIST, which is an arithmetic task where participants must solve problems on a computer screen (see Dedovic et al., 2005 for details). Unlike in the original MIST, here participants believed to be competing against a specific competitor that they met at the start of the session. Specifically, two participants are invited to come-in for each session. In cases where only one participant is available then a stooge stepped-in to create the illusion of a second participant. Participants were welcomed together and shown around the laboratory, seeing where each one would be for the session. Participants were placed in individual rooms, separated by a control room in the middle where the researchers were sat. To increase confidence, participants switched between rooms from the first and the second sessions (De Calheiros Velozo et al., 2021). To avoid any potential anticipation effect, participants were not informed that they would be competing against one-another until the moment where they were to start competing. However, participants were not truly competing with each other as the task was set so that participants had little control over the outcome, only winning around 30% of the time. Moreover, scripted negative feedback was provided to further enhance social-evaluative threat. Participants were told that the goal of the study was to investigate mental effort in order to disguise the true purpose of the study (i.e. stress). We do not report on the task itself as previous work has shown that it was effective in eliciting a significant stress response, with pairwise comparisons showing a significant increase in both perceived stress (PS: $B = 1.67$, $p \leq .01$) and NA ($B = .20$, $p \leq .01$) from baseline to stress. In addition, there was a significant effect of time on both HR and RMSSD with pairwise comparisons showing a significant increase in HR from baseline to stress ($B = 2.26$, $p \leq .01$), and a parallel decrease in RMSSD ($B = -2.08$, $p \leq .01$), see De Calheiros Velozo et al., 2021 for further details.

Experience sampling method (ESM)

The Experience Sampling Method (ESM: see Myin-Germeys et al., 2018 for details) was used to collect daily life data in the four days prior and four days post laboratory stress induction with the rMIST. Participants received ten assessments a day for

eight consecutive days. Assessments were triggered at semi-random times, meaning that they took place in a 15-hour interval during the day with a starting time and ending time clearly established based on each participant's schedule. Assessments occurred at random within this interval, with no less than 15 minutes between them. Participants had 90 seconds to initiate them upon triggering or it would disappear and be counted as missing. Assessments included items on momentary levels of stress, affect, and context measures (Myin-Germeys et al., 2009).

Measures

Psychopathology

Vulnerability was measured with a measure of Psychopathology assessed once at the start of the study using the Symptom Checklist 90 (SCL90: Smits et al., 2015). This instrument includes 90 items assessing several symptoms of psychopathology including agoraphobia, anxiety, depression, somatization, cognitive performance deficits, interpersonal sensitivity, hostility, and sleep difficulty. All items were scored on a 5-point Likert scale from 0.*not at all* to 4.*extremely* with scores ranging from 0 to 360. This measure has shown a high reliability with a Cronbach Alpha of .94.

Momentary negative affect

Participants completed self-report questionnaires assessing momentary measures of affect ten times a day during the ESM period. A measure of NA was computed as the average of the momentary mood items: *worried*, *stressed*, *anxious*, *restless*, *irritated*, and *down*, measured on a 7-point Likert scale from *1.not at all* to *7.very much*. This measure showed a high reliability with a between-person Cronbach alpha of .88 and a within-person alpha of .70. These measures have been used in other studies (Lataster et al., 2013).

Heart rate and heart rate variability

Heart rate, heart rate variability, and accelerometer (ACC) data were continuously collected throughout the eight-day testing period with the Stingray, an ECG-patch from imec that was attached to the chest with a skin-friendly, hypoallergenic, stretchable patch. The Stingray collected ECG sampled at 128 Hz, and motion data (i.e. ACC), which was measured with a 3-axis accelerometer sampled at 32 Hz. The Stingray provides a quality indicator for each datapoint indicating how reliable it is. Values below the confidence threshold were discarded and mean HR, the root mean square of successive differences (RMSSD), and ACC were computed in windows of five-minutes immediately preceding each ESM assessment. The RMSSD is a time-domain measure of heart rate variability measuring the time difference between heartbeats: $\text{RMSSD}=\sqrt{(RRinterval_{n-1}-RRinterval_{n})^2}$ (Shaffer & Ginsberg, 2017).

Skin conductance level

Electrodermal activity (EDA) in daily life was continuously collected with the Chillband, a wearable hypoallergenic wristband from imec. The wristband had a high dynamic range between .05μs and 20μs on the inner side of the wrist (see Smets et al., 2018 for further details on the Chillband). The Chillband provides a quality indicator for each datapoint indicating how reliable it is. Values below the confidence threshold were discarded and the mean skin conductance level (SCL) was computed per windows of five-minutes immediately preceding each ESM assessment.

Event-related stress

In line with other studies, stress was assessed with Event-related Stress (ES: see Lataster et al., 2010). Event-related stress is a subjective measure whereby participants rate the level of stressfulness of the most important event that had occurred since the previous ESM assessment. Events were rated on a Likert-scale from − 3.*very unpleasant* to 3.*very pleasant*. Events scored from 0 to 3 meant that they were not stressful and were therefore recoded to 0. Events scored as −1, −2, and −3, were treated as events where a stressor occurred and were recoded to 1, 2, and 3 respectively, reflecting stressor intensity (Vaessen et al., 2017).

Statistical analytical plan

All analyses were conducted in STATA 14.2, and the statistical analytical plan was delineated and pre-registered on the open science framework prior to data analysis, see https://osf.io/s6rdq/?view_only=bb89e6bfe22046519fddcdb3cb359078 for details (Foster & Deardorff, 2017).

To differentiate the ESM days pre- versus post- laboratory stress induction with the rMIST, a dichotomous measure of Time was computed, with the days that were sampled prior to the stress task coded as 0 and the days following the task coded as 1. Additionally, a variable named Weekend was constructed to reflect days that occurred during the weekday (coded as 0) and weekend days (coded as 1) to control for any potential weekend/weekday effect. Descriptive measures were calculated for all continuous (i.e. means, standard deviations, and ranges) as well as for all categorical (i.e. percentages) variables in the current study.

We conducted multi-level analyses to test whether there were any significant changes in average levels of affect (i.e. NA) and the various physiological parameters (i.e. HR, RMSSD, and SCL) in the four days pre- compared to the four days post- task. Time was used as a predictor and NA, HR, RMSSD, and SCL as outcomes separately.

We conducted similar multi-level analyses to test whether there was any increase in stress reactivity in the days post- compared to pre- task. Reactivity was measured as the effect of Event-related Stress on the outcome measures (i.e. NA, HR, RMSSD, and SCL) (Lataster et al., 2013). Therefore, to measure the effect of the task on daily life stress reactivity we used the interaction between Time and Event-related Stress as the predictor and NA, HR, RMSSD, and SCL as outcomes separately.

Finally, to test whether Psychopathology moderated the association between Time and adverse effects (i.e. change in average levels, and increase in reactivity), we ran two additional models per outcome variable, adding Psychopathology

as an interaction term with: (1) Time, to test if it moderated the effect on average levels and, (2) with Time and Event-related Stress (i.e. three-way interaction), to test if it moderated the effect on stress reactivity.

In all models, individual events (level 1) were nested within subjects (level 2). Additionally, we controlled for age, gender, day of the week (i.e. weekend vs. weekday), and ACC. We allowed for random intercepts of the outcome measure (i.e. NA, HR, RMSSD, and SCL) by subject with AR(1) auto-correlated residuals. We also allowed for random slopes (i.e. Event-related Stress) at level 1 for models testing reactivity. Finally, given the hierarchical structure of the data, all models were fitted using the restricted maximum likelihood estimation (REML).

Results

Descriptive statistics

The sample was comprised of 53 participants, 46 (86%) of which were women. Participants were aged between 19 and 35 years with a mean age of 24 years (SD = 4.0). Over half of participants (60%) were studying at the time of the study and the remaining 40% were working. Participants scored a mean score of 56 (SD = 39) on the measure of Psychopathology: SCL90, with values ranging from 9 to 176. Scores above 70 were understood to be within the clinical range. Here, 14 (26%) participants scored above, and 39 (74%) scored under this threshold (see Table 1).

In the eight days of ESM, a total of 3,565 assessments were completed resulting in an average of 83.75% compliance rate. Thirty-three (62%) participants had the weekend pre-task while the remaining 20 (38%) had it post-task. Of the total number of completed assessments, 3,083 events (86%) were rated as not stressful (i.e. level 0), 325 (9%) as mildly stressful (i.e. level 1) and 157 (5%) as moderately stressful (i.e. level 2). No event was reported as highly stressful (i.e. level 3).

Heart rate and RMSSD were normally distributed, while NA and SCL were right skewed. A log transformation was thus performed and resulted in a normal distribution for SCL but not NA, therefore we maintained the non-transformed measure of NA and the log transformed measure of SCL.

Table 1. Descriptive statistics (i.e. N, mean, SD, min, and max) for outcome measures (i.e. NA, HR, HRV, and SCL), predictors (i.e. Event-related Stress, and Psychopathology).

	N	Mean (SD)	Min-Max
Psychopathology	53	56.28 (38.95)	9 – 176
Event-related Stress (ES)	3565	.18 (.49)	0 – 2
Negative Affect (NA)	3605	1.85 (1.06)	1 – 7
Heart Rate (HR)	2374	78.04 (13.33)	45.76 – 150.03
Root Mean Square of Successive Differences (RMSSD)	2374	83.61 (41.36)	5.29 – 340.99
Skin Conductance Level (SCL)	2578	1.05 (2.25)	.005 – 20.08
Accelerometer (ACC)	3889	.06 (.07)	.001 – .732

Average levels

In the models investigating the effect of Time on average levels of affect and physiology, a significant effect of Time was found for RMSSD and SCL, but not for NA and HR (see Table 2 for results). Participants exhibiting a higher average level of RMSSD and a lower average level of SCL post- task compared to pre-task (see Figure 1). These changes are in the opposite direction from what would be expected based on the stress sensitization hypothesis, suggesting decreased sympathetic and increased parasympathetic activity in the days post-task.

Stress reactivity

Likewise, we tested whether there was a significant increase in reactivity pre- compared to post- task. We found no significant effects for any of the measured parameters (see Table 3). In other words, no stress sensitization was observed in the days following the task.

Moderating role of psychopathology

Average levels

We added Psychopathology as an interaction term with Time to assess whether it was moderator. We found no significant interaction effects for any of the outcome measures, with the exception of NA (see Table 4 for results). Specifically, individuals who reported higher symptoms of psychopathology at baseline were more likely to experience an increase in overall levels of NA post- task compared to pre- task than individuals who reported less symptoms of psychopathology (see Figure 2). On the contrary, no such effects were found for HR, HRV, and SCL.

Reactivity

Finally, the three-way interaction between Time, Event-related Stress, and Psychopathology was not significantly predictive of any of the outcome measures (i.e. NA, HR, RMSSD, and SCL) (see Table 5). Findings indicated that Psychopathology was not a moderator for the association between Time (i.e. pre- and post- task) and reactivity.

Post-hoc exploratory analyses

We found an unexpected increase in average RMSSD and a decrease in average SCL in the days post- task compared to the days pre-task. To further corroborate these findings we conducted two separate paired t-tests to compare the

Table 2. Results from multi-level models with Time and the interaction between Time as the predictor for NA, RMSSD, and SCL separately.

Outcome	Predictor	N	chi2	B	SE	*p*
NA	Time	3605	2185.50	.02	.04	.634
HR	Time	2374	823.70	.09	.52	.864
RMSSD	Time	2373	468.38	3.59	1.79	.044
SCL	Time	2416	945.86	−.42	.13	.001

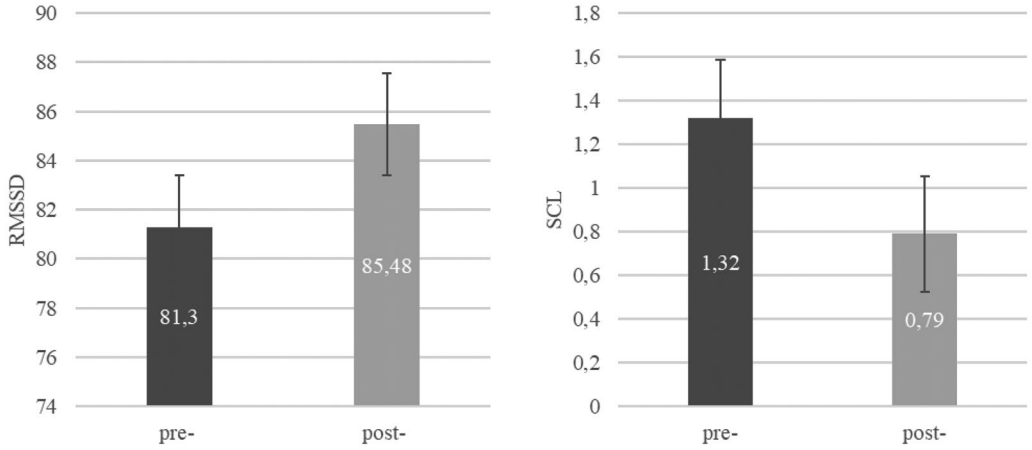


Figure 1. Average levels of RMSSD and SCL pre- and post- rMIST with standard error bars.

Table 3. Results from multi-level models with the two-way interaction between Time and Event-related Stress (ES) as the predictor for NA, RMSSD, and SCL separately.

				ES = 1			ES = 2		
Outcome	Predictor	N	chi2	B	SE	*p*	B	SE	*p*
NA	Time*ES	3565	2180.68	−.147	.085	.084	.197	.126	.119
HR	Time*ES	1906	565.48	−.731	1.442	.612	.613	1.903	.747
RMSSD	Time*ES	1905	300.25	2.300	5.576	.680	−11.441	7.383	.121
SCL	Time*ES	1976	740.78	−.219	.242	.364	.224	.364	.538

Table 4. Results from multi-level models with the interaction between Time and Psychopahotlogy as the predictor for NA, RMSSD, and SCL separately.

Outcome	Predictor	N	chi2	B	SE	*p*
NA	Time*Psychopahotlogy	3605	1761.73	.002	.001	**.042**
HR	Time*Psychopahotlogy	2374	756.46	−.014	.014	.295
RMSSD	Time*Psychopahotlogy	2373	439.75	.033	.046	.475
SCL	Time*Psychopahotlogy	2416	948.52	−.003	.004	.350

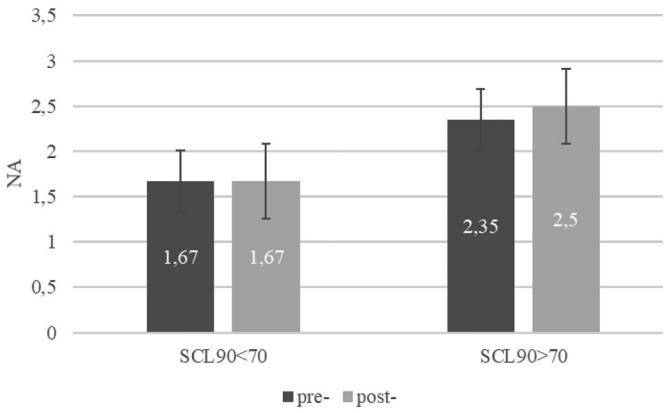


Figure 2. Visualization of average levels of NA pre- and post- task for individuals with lower Psychopahotlogy (i.e. SCL90 < 70) and higher Psychopahotlogy (i.e SCL90.>70) with standard error bars. Scores over 70 are within the clinical range.

average RMSSD and SCL pre- versus post-task. There was a statistically significant difference in average RMSSD pre-task (M = 81.30; SD= 39.49) compared to post-task (M = 85.48; SD= 41.33); with a statistically significant increase of 4.18 ms, $t(2371) = 2.5146$, $p = 0.01$. Likewise there was a statistically significant difference in average SCL pre-task (M = 1.32; SD= 2.56) compared to post-task (M= .79; SD= 1.88); with a statistically significant decrease of .53uS, $t(2576) = -5.9657$, $p < 0.001$.

There is evidence that positive affect (PA) may mediate this association. Concretely, an increase in parasympathetic activity, and a decrease in sympathetic activity during periods of heightened stress may be due to a decrease in PA (Tutunji et al., 2021). To assess whether PA could also explain our results, a measure of PA was computed as the average of the momentary mood items: *relaxed*, *satisfied*, and *cheerful*. Items were measured on a 7-point Likert scale from *1.not at all* to *7.very much*. This measure has been used in other studies (Lataster et al., 2013), and showed a high reliability with a between-person Cronbach alpha of .79, and a within-person alpha of .72.

Simple mediation analyses were performed using the medsem package in STATA (Mehmetoglu, 2018), with RMSSD and SCL as outcomes separately. Time was the predictor variable and PA the mediator. Results from the Zhao, Lynch, & Chen's approach were found to be non-significant for both outcome measures: RMSSD [Effect= 3.75, p=.037, 95% C.I (-.390, .043)], and SCL [Effect= −.562, p=.000, 95% C.I (-.009, .004)]. In short, PA did not mediate the effect of Time on neither RMSSD or SCL.

Discussion

The current study was designed to investigate if a typical psychosocial laboratory stress task (i.e. the rMIST) posed any adverse effect that would linger beyond the laboratory and into daily life. There were no significant adverse changes in daily life average levels of affect and physiology pre- compared to post- task, as well as no increase in reactivity for any of the measured parameters. However, vulnerability was a

Table 5. Results from multi-level models with the three-way interaction between Time, Event-related Stress (ES), and Psychopathology as the predictor for NA, RMSSD, and SCL separately.

				ES = 1			ES = 2		
Outcome	Predictor	N	chi2	B	SE	*p*	B	SE	*p*
NA	Time*ES*Psychopathology	3565	1740.42	−.001	.002	.835	−.002	.003	.471
HR	Time*ES*Psychopathology	1906	522.63	.015	.042	.709	.018	.053	.998
RMSSD	Time*ES*Psychopathology	1905	279.65	.275	.167	.099	.107	.205	.600
SCL	Time*ES*Psychopathology	1976	739.44	.005	.008	.498	−.002	.011	.798

moderator of average NA, in that individuals who reported more symptoms of psychopathology at the start of the study were more likely to experience an increase in average levels of NA in the days that followed the task. Finally, there was an unexpected increase in average RMSSD and a decrease in SCL pre- to post- laboratory stress induction.

Daily life adverse effects

There were no daily life adverse effects ensuing from the task, in terms of average levels and reactivity. These findings align with what is known about stress-sensitization, which is most likely to occur following a major life stressor (e.g. trauma) or repeated/prolonged exposure to stress (Harkness et al., 2015). As such, a single mild to moderate stressor, as is the case of the rMIST, should not be sufficient, on its own, to elicit significant changes to the stress system. Consequently, findings provide support for the use of the rMIST to investigate the various parameters of the stress response without posing a significant risk to participants, especially less vulnerable participants who do not exhibit many symptoms of psychopathology.

Participants who reported higher levels of psychopathology were more likely to experience an increase in NA in the days following the stress task compared to individuals who reported lower levels of psychopathology. These findings are in line with our hypothesis, in that individuals who are already more stress-sensitive are likely to experience mild stimuli as more stressful than individuals who are less stress-sensitive, resulting in more sustained effects (Bale, 2006; Guidi et al., 2021; Krkovic et al., 2018). In fact, it has been shown that these individuals take significantly longer to recover from daily stressors compared to healthy controls (Vaessen et al., 2019). Consequently, it may be that they are unable to entirely recover from the stressor in the days following the task.

In addition, psychopathology is associated with behaviors known to prolong negative affective states, such as typical methods of perseverative cognition like worry and rumination (Brosschot et al., 2006; Van den Bergh et al., 2021), which can exacerbate stress-sensitization (Ruscio et al., 2015). As such, the affective response to the stressor can persist even after the stressor has passed. Moreover, individuals who engage in more perseverative cognition have shown to both experience stress more intensely with heightened reactivity and longer recovery periods (Capobianco et al., 2018; Verkuil et al., 2010). Together, these processes may be driving the effects of the task to linger in individuals who are more vulnerable and at a greater clinical risk.

It should be noted that the effect of Psychopathology is only weak, and therefore we cannot draw firm conclusions. A replication study on a larger sample of individuals with heightened vulnerability would be necessary to reinforce our findings. For now, based on current findings, we conclude that caution should be applied when using the task in more vulnerable individuals. Precautionary measures such as providing a more extensive debriefing following the rMIST may help minimize this effect (Benham, 2008).

Physiological unwinding post rMIST

The observed changes in RMSSD and SCL show a physiological unwinding following the rMIST that lingers for several days following the task. These findings confirm the idea that physiological arousal elicited by the task may not persist beyond the task itself, albeit the reversed pattern is notable. It is known that the body naturally relaxes following a period of high expenditure (Sonnentag et al., 2017; Zijlstra & Sonnentag, 2006), such as engaging in a task where stress is elicited. This is the body's way of recovering from a stressor and is necessary to avoid allostatic overload (Geurts & Sonnentag, 2006; Sonnentag et al., 2017). Longer periods of recovery are especially necessary following more chronic or enduring stress like work stress (Geurts & Sonnentag, 2006). However, the rMIST is an acute stressor as opposed to the more chronic stress associated with work. As such, physiological unwinding would be expected to linger for several hours following the stressor but probably not days. Therefore, recovery may not entirely explain the observed physiological unwinding.

A potential explanation for our findings was that individuals may have experienced less PA in the days following the rMIST that would translate to less physiological arousal. In fact, in a separate study, PA was found to mediate physiological changes echoing our findings. Namely, daily life measures of sympathetic activity (i.e. HR and SCL) were significantly lower during a week of increased stress (i.e. stress week) compared to a week of lesser stress (i.e. control week). The decrease in physiological arousal observed during the stress week was explained by a decrease in PA instead of an increase in NA (Tutunji et al., 2021). It is known that pleasant events, and even the mere anticipation of pleasant events, can elicit a physiological response comparable to that of unpleasant events, with a significant increase in sympathetic activity and a decrease in parasympathetic activity (Löw et al., 2008). Thereby, the experience of stress may reduce the experience of positive events (Rackoff & Newman, 2020), resulting in an overall decrease in physiological arousal. A similar mechanism could have been at play here as well, wherein participants could have experienced less emotions of positive valence in the days following the rMIST compared to the days prior resulting in the observed changes. However, exploratory

analyses did not provide support for this theory. Specifically, PA did not appear to be a mediator and therefore could not explain the measured effects. Therefore, it is likely that another factor may be driving these effects.

On the contrary, it is also possible that the physiological unwinding actually reflects a return to normal following several days of heightened physiological arousal in the anticipation of the laboratory task. In fact, it has been shown that the mere anticipation of a stressor can be equally as stressful as the stressor itself (Brosschot et al., 2006; Capobianco et al., 2018). Moreoever, these effects can linger (Neubauer et al., 2018). Although participants were not aware that they would be subjected to a task designed specifically to elicit stress, the mere prospect of participating in a laboratory task may have been stressful in and of itself.

Limitations

Findings should be viewed in light of several limitations. First, the sample was comprised of volunteers recruited from the general population; that is, individuals who did not report any significant health complaints prior to their participation in this study. More specifically, participants had not received any clinical diagnosis or treatment for a mental illness. Although findings are meaningful, they would benefit from being replicated directly in a larger sample with symptoms of psychopathology within the clinical range.

Secondly, we could not explain the observed physiological changes in RMSSD and SCL. Although it is not expected that these changes be adverse, they nevertheless indicate that something else may be happening that we are not cognizant of. A follow-up study may prove useful in identifying additional potential adverse effects that were not accounted for in the current study.

Thirdly, aside from the decrease in average levels of SCL, findings are moderately weak. In fact, should a Bonferroni correction be performed the observed change in RMSSD and the moderating role of Psychopathology would seize to be significant. Although it is expected that these effects are meaningful, they may not suffice to formulate strong conclusions. A replication study in a larger sample is necessary to assess whether these effects remain.

Finally, repeated measures naturally introduce noise and complexity to the data. As such, it is important to consider the added risk of it diluting or obscuring potential effects. Despite this risk, mixed models are especially suited to handle these data and, when applied correctly, should suffice to yield reliable findings.

Conclusion

We tested whether a typical psychosocial laboratory stress task would pose a significant daily life adverse effect on participants. Although the task generally did not pose any significant adverse effect, more vulnerable individuals, were more likely to experience sustained NA in the days that followed the task. However, given that these effects were weak, replication studies are needed to solidify this finding. Finally, unexpected changes in average levels of RMSSD and SCL indicate that there may be another, still unidentified, factor driving these changes.

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Disclosure statement

All authors certify that they have no financial or personal conflict of interest to disclose.

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Data availability statement

All analyses have been pre-registered prior to data analyses. Data is available upon request.

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